

THEJUS MAHAJAN PH. D.

Bioinformatics & Data Scientist

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Hamburg, Germany



CAREER INTERESTS

Computational Scientist (PhD) with 5+ years of experience in **Biological Systems Modeling** and **High-Dimensional Data Analysis**. Experienced during post-doc in applying **stochastic population models** in hindcast and forecast projections. Trained in **Python and R pipelines** for noicy data analysis. Seeking for a career shift to Bioinformatics. I am leveraging on my recent Bioinformatics and Biostatistics training along with the advanced mathematical modeling expertise gained during the Ph. D. and post-doc period to solve problems in **human metabolic networks** and clinical data analysis.

EXPERIENCE

Further Training - Application-related Bioinformatics and Biostatistics

CQ Beratung + Bildung

08/2025 - Present | Berlin, Germany

Training in Bioinformatics and Biostatistics

- Introduction to programming and Python, including Shell (BASH), programming methodology, and object-oriented programming in Python.
- Bioinformatics and databases, including NCBI, Biopython, web services, and database management with SQL.
- Applied bioinformatics, including sequence analysis and strategies, structural bioinformatics, software for structure visualization, and NGS analysis using Nexflow and Galaxy.
- Applied biostatistics, including statistical analysis using R, descriptive and multivariate statistics, ANOVA, PCA, HCA, workflows using R/Bioconductor, and Markov chain implementation.

Guest Scientist

Helmholtz-Zentrum Hereon, Ökosystemmodellierung

05/2025 - 10/2025 | Geesthacht, Germany

Computational Biologist & Systems Modeler

- Spearheaded a complex research project modeling evolutionary processes in marine environments, in collaboration with the Ecosystem Modeling Department.
- Applied advanced computational techniques to simulate the long-term impacts of environmental changes on marine ecosystems.

Post-doctoral Scientist

University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences

08/2021 - 01/2025 | Hamburg, Germany

Marine Ecosystem Modelling

- Model development: Developed the novel "Cyanobacteria Life Cycle" (CLC) model within the ERGOM framework to project future climate warming scenarios.
- High-performance computing: Engineered models in Fortran and analyzed large NetCDF datasets using Python in a Linux environment.

Ph.D. Researcher in Astrochemistry

University of Paris-Saclay, France

10/2015 - 09/2018 | Orsay, France

Led atomic collision experiments, analyzed data, and modeled outcomes.

- Big Data Analysis: Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- Algorithm Development: Utilized C++ for noise cancellation, signal smoothing, and converting raw data into multidimensional structures.
- Simulation: Developed a molecular fragmentation model to predict chemical reactions in space, contributing results to the KIDA database.

STRENGTHS

★ Advanced Scientific Modelling

Developed and implemented a novel biogeochemical model (CLC) within the GOTM-FABM framework to run complex climate change projection scenarios.

💡 Complex Problem-Solving

Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector, successfully recalculating experimental results using multi-sensor data.

🔗 Bioinformatics & Biostatistics

Successfully bridged physical science expertise with modern biological workflows. Skilled in NGS data analysis, sequence alignment, and multivariate statistical modeling using R (Bioconductor) and Python.

TECHNICAL SKILLS

Programming

Python (Pandas, NumPy, Scikit-learn), R (Bioconductor, ggplot2), Fortran, SQL, BASH

Bioinformatics Tools

NGS Analysis, Galaxy, Sequence Alignment, Structural Bioinformatics

Biostatistics

Multivariate Analysis, PCA, ANOVA, Dimensionality Reduction

LANGUAGES

English

Proficient



German

B1 - Goethe Certified



French

Intermediate



Malayalam

Native



Tamil

Proficient



Hindi

Intermediate



Sanskrit

Intermediate



EDUCATION

Ph.D. in Astrochemistry

Institute of Molecular Sciences (ISMO) Orsay, Université Paris-Saclay, France

📅 2015 - 2018 📍 Orsay, France

M.Sc. in Physics

National Institute of Technology Calicut

📅 07/2012 - 01/2015 📍 Calicut, India

REFERENCES

Prof. Dr. Kai Wirtz

kai.wirtz@hereon.de

Prof. Dr. Elisa Schaum

elisa.schaum@uni-hamburg.de

TRAINING / COURSES

High Performance Computing (HPC) Training

Jülich Supercomputing **Center (JSC)**,
Forschungszentrum Jülich

MITx: 7.00x Introduction to Biology - The Secret of Life

Comprehensive coursework on genetics, biochemistry, and molecular biology.

PUBLICATIONS

🔗 <https://scholar.google.com/citations?hl=en&user=PJkZwAwAAAAJ>